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*ILLUSTRATED WITH TWENTY-TWO MAPS.*

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## THE SEAT OF WAR.

THE accompanying Map of the Seat of War covers the ground which is likely to be the chief centre of interest during the present autumn, and perhaps throughout the winter. It includes the whole of the wedge-shaped region to the south of the Danube, now occupied by the Russian army. If the Czarewitch succeeds in maintaining his position in front of Biela, and along the line of the River Jantra, the country which will probably be occupied during the winter can already be defined. It is of triangular form, with the Danube as its base from the mouth of the Osma at Nicopolis to the mouth of the Jantra 20 miles below Sistova. The two sides are formed by the Rivers Osma and Jantra. The former, rising in the Balkans between Slatitza and Shipka, flows north to Lovatz, then takes a wide sweep to the eastward, finally turning north-west, and falling into the Danube at Nicopolis, after a course of 70 miles. To the westward, in the valley of the Vid, is the strongly-entrenched position of Plevna, still held by the Turks. The river Jantra, the eastern side of the triangle, rises in the Shipka Pass, flows northwards past Gabrova and Tirnova, the ancient Bulgarian capital, to Biela, and thence, taking a sharp turn to north-west, continues on that bearing to the Danube. To the eastward of the Jantra are the valleys of the Black and White Lom, with Rustchuk at the mouth of their united stream, and Rasgrad on the banks of the White Lom. Shumla is within the basin of the river flowing into the Black Sea at Varna; and this eastern portion of Bulgaria, from the Jantra to the sea, is occupied by the largest of the Turkish armies, commanded by a renegade German. Thus the Russian position is in the form of a wedge driven into the heart of Bulgaria, with its apex in the Balkan mountains. It cuts the Turkish forces in half, the larger army being to the east of the Jantra, and the smaller to the westward of the Osma, in the extemporised fortress of Plevna. A third Turkish army threatens the apex of the triangle in the famous Shipka gorge: and it is here that the most heroic deeds as yet recorded have been performed by the Russian defenders of the pass.

The chain of the Balkans is not connected with the Illyrian Alps, but is a distinct system, extending from the valley of the Timok which separates Servia from Bulgaria, to the Black Sea. The Balkans thus form a vast and elevated plateau, sloping gently down towards the Lower Danube, but with an abrupt descent to the south. From the Black Sea to Slivno the steep southern slopes are composed of cretaceous rocks, intersected by porphyries. To the west of Slivno these slopes are of granite and gneiss, while between Shipka and Slatitza mica-schists and clay slate predominate. The western portion of the range, bounding the basin of Sofia on the north, consists entirely of Triassic sandstone and limestone. In the copy of Hochstetter's Geological Map of Eastern Turkey, given in our number for November 1873 (facing page 328), there is a section of the Balkans a little to the eastward of Shipka, showing the lofty summit immediately above the convent of Mikhin, and another from Trevna to Eski Zagra. A third and most instructive section, on the same map, is from Nicopolis, on the Danube, by Plevna, and across the Balkan range at the Iskor defile, to the basin of Sofia. Here are shown the pliocene diluvium of the valley of the Danube, the miocene deposits of the Osma and the Vid, the upper chalk and gault, the caprotina limestone of the bases, and the Neocomian limestone of the summits, in that part of the Balkan range.

The Shipka region of the Balkan is within the crystalline zone. The strata are much contorted, and dip generally 80° south, containing many quartz reefs; and near Kazanlik granite and gneiss are found. The Shipka Pass is almost in the centre of the Balkans, half-way between the valley of the Timok and the Black Sea. From the north it is approached by following the river Jantra to Gabrova, a town inhabited by industrious Bulgarians, and consisting of about 1,500 houses. Thence the road ascends, through a magnificent forest, to the actual pass, which is 4,314 feet above the level of the sea. Then the first view is obtained of the plain to the south, the famous "Valley of Roses" of Kazanlik. The steep slopes and profound valleys are streaked here and there with forests of oak and beech. The route descends from the pass very rapidly, and, while the road from Gabrova to the summit is reckoned at 4½ leagues, the village of Shipka is reached in an hour after leaving the highest point. Shipka is a village with about 800 Bulgarian houses and two churches, and is situated in a gorge bordered by forests. Another hour is sufficient to reach Haskioi, in the plain, and another brings the traveller, through walnut groves and rose gardens, to Kazanlik, a town with 21,000 inhabitants.

The Shipka Pass is the apex of the triangle occupied by the Russian army; and its gorge, as well

this province for agricultural purposes, with greater fertility of soil, cultivated to the highest degree by skilful labourers, who have made it the "Garden of the Provinces," and the Yangtsze Kiang the "Girdle of China." SAMUEL MOSSMAN.

## A DESCRIPTION OF THE ISLAND OF FORMOSA.

### WITH SOME REMARKS ON ITS PAST HISTORY, ITS PRESENT CONDITION, AND ITS FUTURE PROSPECTS.

IT is only a very few years since the Island of Formosa, or Taiwan, as it is called by the Chinese, was practically an unknown land, and even now the number of Europeans who have penetrated into the interior is very small.

In April 1877, business took me to the island, and during my stay I visited all the treaty ports, and journeyed overland from Taiwan-fu to Kelung, a distance by road of 218 miles.

My business gave me the *entrée* of the Governor's Yamen, and required that I should collect as much information as possible, from observation and from reports, as well as from interviews with those acquainted with the country. Although I did not penetrate into the savage district, and cannot therefore describe any previously unknown land, still I believe that much of the information contained in the following pages has never been published, and I have no doubt will be found useful by any one wishing to visit the island, or to make himself acquainted with its present condition or future prospects.

Formosa is an island about 240 English statute miles long and 90 miles wide, lying between N. lat. 22° and 25° 20′, and about 120 miles distant from the coast of China.

The Chinese had a knowledge of the existence of the island at a very early date, but they do not appear to have become really acquainted with it until the year 1430, when some junks were driven there by stress of weather. It is probable that most of the very early Chinese settlers were shipwrecked sailors.

About the year 1622, the Dutch settled in the Pescadores, a group of small islands lying about 30 miles to the west of Formosa. In 1624, at the instance of the Chinese, they removed to Formosa, and there built Fort Zelandia, the ruins of which still exist near Taiwan-fu.

In 1626, the Spaniards built a fort at Kelung, but the Dutch drove them out, and, to make themselves more secure in the north of the island, they built a fort at Tamsui, part of which now forms the office of the British Consulate.

In 1644, the Manchu Tatars gained Pekin, and very soon afterwards became rulers of the greater portion of China, and, in order to escape from the conquerers, 25,000 Chinese emigrated to Formosa.

In 1662, a pirate chief called Koxinga, at the head of a force of Chinamen, drove out the Dutch and became ruler of the island; and in 1682 Formosa passed into the hands of the Imperial Government of China, and it has ever since remained part of the Chinese province of Tokien.

Although nominally a part of China, the sovereignty of the Chinese only extends over about half the island; over certain other parts they exercise a nominal sway, but there are still some portions into which no Chinaman dares to enter.

Formosa has not been without its troubles. In 1782 it suffered from a disastrous inundation, and in 1788 there was a serious rebellion. Again, in 1832 to 1833, there was a rebellion caused by the oppression of the Chinese officials. The horrors of this rebellion were aggravated by the fact that the various clans of Chinese who had emigrated to Formosa from different parts of China, took this opportunity of fighting with each other. This division of the camp, however, enabled the authorities eventually to quell the insurrection.

Up to much more recent times the island was noted principally as a place where a shipwrecked crew were sure to be murdered, and likely to be eaten. Although the Chinese inhabitants were never accused of being cannibals, they were quite as desperate wreckers as the natives.

One of the most noted cases was that of the British brig 'Ann,' wrecked on the coast of the Tamsui district on the 11th March 1842. The crew and one passenger, fifty-seven in all, reached the shore, and tried to get away in two junks. The people on shore, however, would not allow this, and took them all prisoners. They were stripped naked and put in prison. They were soon marched to Taiwan-fu, having no clothing with the exception of a few articles given them by the more kind-hearted of the Chinese. Here they were imprisoned in a granary. On 13th August forty-four of them were taken to the execution ground, and forty-three beheaded. One man, having obtained a little money, had bought some drink and became intoxicated. He stood on his head before the magistrate and muttered some words he had heard used in a joss-house. He then performed some acrobatic feats, and the magistrate said he was a good man, and should not be beheaded. He was afterwards liberated, with seven others, and arrived in Amoy on 24th November 1842. One other was sent to Hong Kong, where he arrived 2nd April, 1843; two died on the day of the wreck; two died in prison; and one escaped. At the same time a large number of men from the 'Nerbudda' were murdered. The total number of British subjects beheaded, including those from the 'Ann,' was 197.

On the matter being investigated, it appeared that the Chinese officials in Formosa represented to the authorities at Pekin that these ships were war vessels, and had been captured after a severe battle; and not only did the Central Government approve of beheading the prisoners, but the officials received promotion. When the British Government found out the truth, the matter was brought before the notice of the Emperor, and a proclamation was issued degrading the Formosan officials, and ordering the principal ones to be brought before the Board of Punishment. It does not appear that anything further was done. From time to time, after this, wrecks took place on the Formosan coast, and, in each case, all or some of the crew were murdered.

In September 1850 the 'Larpent' was lost on the south coast, and only three of the crew escaped; some of the remainder being drowned, and the others murdered.

In 1851, when a Chinaman was sent to Formosa to search for any sailors who might be in captivity, he was shown a warrant for the arrest of all foreigners, and of all Chinese in the service of foreigners, who were found on the island.

Matters remained in this position till 1858, when Taiwan-fu was opened to foreign trade by the British, French, and United States treaties, and Tamsui by the French treaty in the same year.

In 1863 Takao was opened as a dependency of Taiwan-fu, and Kelung as a dependency of Tamsui, by provisional Customs regulations.

The principal Customs Office in the north of the island is at Tamsui; but in the south of the island it is at Takao, the office at Taiwan-fu being subordinate to it.

British Consuls were first appointed to Formosan ports in 1862. The trade of the island was so small that in 1868 it was proposed to take them away, and so save the expense of the consular establishment. The merchants, however, protested, and explained that the trade was small only because it was not developed, and that, after leading them to suppose that consuls would be stationed permanently, and so inducing them to invest capital in Formosan trade, it was most unfair to remove them while the trade was promising to increase. As late as 1871, reference is made in the consular reports to the great excitement the proposal had occasioned. Trade has now increased to such an extraordinary extent that there is no chance of the proposal being repeated.

Even subsequent to the appointment of British consuls, the natives were in the habit of murdering the crews of any vessels wrecked on their coast. The crew of the U.S. vessel 'Rover' met with this fate, and the U.S. consul at Amoy, of whose district Formosa formed a part, seeing that it was useless at that time to negotiate with the Chinese authorities, who were powerless, determined to go direct to the savages.

On the 15th October, 1867, this gentleman, General Le Gendre, made a treaty with Tok-e-Tok, the chief of the savage tribe inhabiting the southern portion of the island. This chief undertook, on behalf of his tribe, to abstain from molesting strangers, and promised to supply them with water and provisions, if necessary; and the Chinese of the district undertook that if any stranger fell a victim, they would search out the guilty parties and punish them. This treaty, although not entirely successful, has had most beneficial effects. Several sailors wrecked since then, on the southern coast, have been sent to foreign settlements.

General Le Gendre was one of the first who did anything towards the topography of the interior of the island.

Some few years ago, some inhabitants of the Lew Chew Islands, over which the Japanese claim sovereignty, were murdered by the savages in Formosa. In order to avenge their death, and to prevent a repetition of such barbarities, the Japanese sent an expedition, in 1874, to punish the culprits. The Chinese disclaimed responsibility; and when, in consequence of this the Japanese landed their troops, the Chinese were pleased and saluted them. After some time, however, they became uneasy, and tried to drive out the Japanese; and failing in this, tried to make some compromise. Eventually, through the medium of Mr. (now Sir T.) Wade, it was arranged, on 6th November, 1874, that the Chinese should pay 100,000 taels, as compensation to the families of the Japanese subjects who had been killed, and 400,000 taels for roads made by the Japanese soldiers —in all, 500,000 taels.* It was agreed that all the Japanese should leave the island before 20th December, 1874. For the part Sir T. Wade took in these negotiations he received the thanks of the British, Chinese, and Japanese Governments.

In 1862 a Spanish vessel was wrecked on the coast, and the cargo was plundered. The Spaniards are now making a claim for compensation. As negotiations are still proceeding, all the facts have not been made public; but as far as one can judge, the demand seems to be a most preposterous one, as the portion of the island where the vessel was wrecked was, at the time, in open insurrection, and all the officials have been punished; and if the claim was to have been made at all, it should not have been delayed fifteen years.

Much interesting information respecting Formosa may be obtained in the Chinese repository, in a work on the Treaty Ports, by Mr. Mayers, and in the account of a cruise by H.M.S. 'Dwarf,' by Captain Bax.

Having now given an outline of the history of Formosa, it will be convenient to give some account of the Treaty Ports and a few of the more important cities, before proceeding to speak of the present condition and future prospects of the island.

One of the ports first opened to foreigners was Taiwan-fu, the capital of the island. This is a large city, situated about 3 miles from the coast. It contains over 120,000 inhabitants. The walls are about 20 feet high, 12 feet wide, and 6 miles in circuit. Until lately they were in a very wretched condition, but they have now been repaired. Taiwan-fu is the seat of Government of the island. The Taotai of this city is said to have the power of life and death, which is possessed by no other Taotai in China. The port of Taiwan-fu is Amping. This is a small village, situated close to the ruins of the old Dutch fort, Zelandia, the greater part of it being built of bricks from the old walls of the fort. There are barracks here for a considerable number of troops. A new fort has been built in the neighbourhood, armed with heavy breech-loading guns; this fort, unfortunately, I was unable to visit. The custom-house and one foreign-office are situated at Amping, but the foreign hongs are at Taiwan-fu. A creek runs up to Taiwan-fu, so that cargo boats can go right up to the city at high tide. The land between Taiwan-fu and Amping is a mud flat, below the level of extreme high water. This flat is divided into squares by embankments, and there is an embanked road all the way. At certain seasons the sea is admitted to this flat, and allowed to evaporate, for the purpose of collecting the salt. The manufacture of salt is a Government monopoly, and smuggling salt is punishable by the heaviest penalties. The salt supplied in Formosa is wretched, being of a dark grey colour.

* There are many different taels in China, and the values of them, compared with English money, are constantly varying. For rough calculations, three Hai Kwan Custom House taels are taken equal to 1*l*.

The harbour, if such it may be called, is only an open roadstead. Vessels anchor in 5 or 6 fathoms 2 miles from the shore. During the N.E. monsoon, *i.e.* from December to March, the roadstead is tolerably well sheltered, and even during November and April there is little to fear. During the S.W. monsoon, however, it is very dangerous.

Close to Amping there is a little deep water; but there is a bar with only 2 or 3 feet of water, and, with the slightest sea, the surf begins to break on it, so that there are often days during which no communication can be held between the shore and the ships. All the trade, consequently, is carried on during the five winter months. Cargo is carried in large boats, but passengers are taken in catamarans. A catamaran consists of twelve bamboos, placed side by side, lashed together with rattans, and small bamboos laid across; two bamboos are lashed a little higher than the others, and to these rowlocks are attached; a square block of wood is lashed near the middle, into which a mast can be stepped. Each catamaran is managed by three men, who row standing, and facing the bow. These catamarans are very safe, and rarely upset on the bar; but they do not venture out in the worst weather. The water, of course, surges up and down between the bamboos, and, in order to keep passengers dry, a tub is lashed on, large enough to hold two people. Fishermen proceed many miles to sea on these apparently frail rafts.

The chief export from Taiwan-fu is sugar. This is grown all over the plain of Formosa, and is crushed in the rude mills of the country. The mill consists of two nearly smooth stone rollers, about 3 feet diam. and 3½ feet high, set vertical and geared together. To the axle of one a long lever is attached, to which two buffaloes are yoked. The cam is fed by hand, and the juice runs through a pipe to a building close by, where it is boiled, passing through from three to six pans, according to the ideas of the owner. The resulting sugar is somewhat dark-coloured, and contains a large quantity of lime, too much being added during manufacture. It is packed in baskets, containing about 2 cwt., and sent to Amping, where it is weighed and re-packed in bags, tied with rattans, each bag containing 1 picul, or 133⅓ lbs. The cane in Formosa is passed twice through the rollers, and each mill crushes enough cane to give, on an average, 15 cwt. of sugar per day.

Another article of export at Taiwan-fu is rattan canes. These grow in large quantities in the island, and are very long and strong, but of a bad colour. They are not considered good enough for putting round tea-boxes, although they are used for sugar-bags. Light-coloured rattans are, therefore, imported from the Straits of Malacca, while dark-coloured and, therefore, cheaper canes are exported to the same place.

About 27 miles to the south of Taiwan-fu lies the port of Takao. Although this port was, as already stated, opened as a dependency of Taiwan-fu, it is now the head-quarters of the Customs staff. The harbour is better than that of Taiwan-fu (indeed, it could hardly be worse); but, as the latter is the capital of the island, ships go there in preference to Takao, when they have a choice. The harbour of Takao consists of a large lagoon, 2 miles long, and 1 to 2 miles wide. The area of water available for large ships is small; indeed, there is hardly room to swing in 3 fathoms, but one or two vessels may lie moored stern and stern. There is a considerable area of 12 feet water. The harbour is capable of improvement. The expense would of course be considerable, as is the case in all harbour works; but the improvement, if properly carried out, would be lasting.

The entrance to the harbour is only 200 feet wide, between steep rocks. The water in the entrance is deep, but outside there is a bar, with 7 to 9 feet water in the centre, and 10 feet at the ends. Large vessels do not, as a rule, come into the harbour, but lie outside, where there is good anchorage in the N.E. monsoon; but the roadstead is dangerous in summer. Immediately to the north of the entrance, Ape Hill is situated. This is a conical hill of coraline stone, 1100 feet high. A good many apes are to be found on it. In the neighbourhood of Takao there are plenty of snipe, but not much other game. The articles of export from Takao are the same as from Taiwan-fu, but less in quantity. In the Customs reports the two ports are treated as one. Lately the British Consul was stationed at Takao; but it is proposed, for the future, that he should live at Taiwan-fu.

In the north of the island the principal port is Tamsui. The real name of the port is Howei, or Hobay, Tamsui being the name of the district of which Teukcham is the principal city. Now, however, the port is known to the Chinese by the name of Tamsui. At the entrance to the harbour there is a bar a mile wide, with only seven or eight feet at low water; vessels, therefore, can only enter at high tide. Inside there is room for several vessels in 12 feet or 15 feet water. The harbour is, on the whole, a good one for its size.

Tamsui is the head-quarters of the Consular and Customs establishments for the north of Formosa. The British Consular offices are situated in the remains of the old Dutch fort, which have been repaired and fitted up for the purpose. About 12 miles up the river is situated the city of Banka, or Meng-ka.

Banka is the largest town in the north of the island, and contains 30,000 inhabitants. Here all the trade of the district is centred. All the merchants interested in the export of tea or camphor, or in the import of opium or piece goods, live in Banka.

Although the foreign settlement is at Tamsui, at the mouth of the river, the British merchants have found it necessary to establish a subsidiary settlement at a place called Twa-ta-tia, about a mile from Banka. Here they have hongs (or combined dwelling houses and offices) and godowns (or warehouses), where tea and camphor are stored, ready to be sent off in cargo-boats, for shipment at Tamsui.

No boat drawing more than 2 feet can go as high as Twa-ta-tia at low water, but there is some deeper water at Banka, and junks go there at high tide. They often ground, however, at low water. The principal articles of export at Tamsui are tea and camphor in foreign ships, and indigo and rice in junks.

Contrary to the practice on the mainland of China, almost all the tea is sold to the foreign merchants, after it has been sun-dried only. The growers sell direct to the foreigners, in small lots, and the taster, after judging of the qualities, mixes those which he considers to be of the same description, and so forms

one chop. On the mainland the tea is bought up in small quantities from the growers by Chinese dealers, who fire and mix the tea before selling it to the foreign merchants.

After the Formosan tea has been bought by the foreign merchant, it is fired under foreign supervision. This is done over holes lined with brickwork and filled with charcoal. After the charcoal has burned down, it is covered with ashes, and remains hot for many days. The tea is placed in baskets over the holes, and remains there three or four hours. The firemen are Chinese, who come over from the mainland for the season. The number who come yearly is now over 2000, and is rapidly increasing. The Chinese merchants are anxious to get the firing into their own hands, and are making a hard struggle for it. According to the reports of the customs officials, they have a good chance. Most of the tea is of the description known as Oolong, and goes to the American market.

The construction of tea-boxes is an industry of considerable importance. The wood comes from Foochow, ready cut to size, but the boxes are put together, covered with paper, and lined with lead in Twa-ta-tia. In 1875 300 tons of lead were imported, the whole of which was used for lining tea and camphor-boxes. The tea is packed in chests, half chests, quarter chests, &c. A chest weighs 105 pounds, and contains 90 pounds of tea.

Next in importance to tea comes camphor. This is obtained from the camphor trees near Tokoham. The root of the tree is cut into chips, from which the camphor is distilled. In most cases the remainder of the tree is allowed to lie and rot. In cases where there are any means of transport the tree is sawn into planks, 7 feet long. 2 feet broad, and 1½ to 3 feet thick. These are worth, at Tamsui, about 25c. per cubic foot. The thickness is gradually increasing, because the freight is usually so much per plank. After the camphor is distilled, it is packed in tubs containing about 4 cwt., and sent to Banka or Twa-ta-tia. It has the appearance of slightly dirty snow, and, in the winter of 1875 to 1876, some camphor, largely adulterated with this article, was bought by some Chinese merchants. On arrival at Twa-ta-tia camphor belonging to foreigners is pressed, with a loss of about 20 per cent. in weight. The water which runs from the press contains a certain amount of camphor oil, which is worth a small amount of money. Screw presses have been used up to the present time, but an hydraulic press is now being erected (April, 1877). The Chinese, having no means of pressing, ship the crude camphor in tubs to Hong Kong. The loss of weight in the passage is 30 per cent. The loss in the case of pressed camphor is very trifling, and as the loss in pressing consists chiefly of water, while the loss on the passage represents to a considerable extent a loss of camphor, the advantage of pressing is very great, not to speak of the saving of freight.

A small amount of sugar is exported from Tamsui, but the quantity is very trifling compared with that exported from the south of the island.

Within 8 or 10 miles of Tamsui there are large sulphur springs, which discharge a considerable quantity of sulphur. The incrustations round the springs are gathered and melted in large pans, something like sugar pans. The scum and the sediment are rejected, and the purer portion is ladled into wooden moulds. The blocks thus obtained are remitted in the same way, in order to obtain somewhat purer blocks. The resulting blocks are of a tolerably good colour, but the whole process is very rude, and while it gets rid of a certain amount of impurities, it introduces some others. The system is less wasteful, but less effective, than the Sicilian method of burning in large heaps or calcarone, and much inferior to the processes employed at a few of the best-managed Sicilian mines. The sulphur is worked by the Government solely for the purpose of powder manufacture, and there is little probability of their ever allowing it to be worked for any other purpose, or by any other persons.

Near Tamsui there are some flour mills, the machinery of which is turned by water-wheels. The usual power used for this purpose in China is either manual labour or bullocks.

The neighbourhood of Banka is the chief tea-producing district in Formosa. Tea is grown on the slopes of the hills in small patches. The scenery is very picturesque. The hills are not very high, and the valleys are small, and there is an utter absence of the grandeur of other parts of the island; but the woods and the varied cultivation have a pleasing effect. On the level ground little is grown but rice. This crop requires a great deal of water, and it is a great advantage to get a stream at a sufficiently high level to irrigate the land by gravitation. There is one plain near Banka of 30 or 40 square miles in area, which is irrigated by water from the hills, which are situated at a considerable distance. This water is carried across the river about 4 miles above Banka by an aqueduct. The supply is said never to fail.

Although in this district, as well as in other parts of Formosa, water at a high level is often available, still it not unfrequently happens that more or less extensive areas have to be irrigated from streams below the level of the ground to be flooded. In such cases different expedients are adopted, according to circumstances. The commonest method is the use of a form of chain pump—a trough open along the top is laid at an inclination of about 1 in 2. A wooden endless chain, with paddles attached to fit the trough, is passed over axles at top and bottom of the trough. The bottom of the trough dips into the stream or pool, while the top overhangs the irrigation ditch. The top axle is made to revolve, so as to make the chain with the paddles attached draw the water up the trough. The loss of power through leakage is not very great. The top axle is made to revolve in one of three ways—by cranks turned by hand, by an arrangement like a treadmill turned by the feet, or by a bullock turning a horizontal wheel geared into the axle.

When the water is to be raised from a running stream, advantage is taken of the power of the stream in a very ingenious manner. A large-framed wheel of bamboo is arranged so as to dip into the stream, the current of which makes it revolve. On the rim of the wheel, pieces of bamboo are fixed at a slight angle, each of these is filled with water as it passes through the stream, and as it comes to the top it discharges the water into a trough, which leads to the irrigating canal. These wheels seem to require very little attention. Advantage is also taken of running

water to obtain power for hulling rice. This is generally done by hand power or foot power, and is very hard work. Where a stream is available, the hammer is arranged with a bucket at the end of the handle. A small stream of water is brought so as to discharge into this bucket. The handle of the hammer being hung on an axle, the weight of water in the bucket lowers the bucket and raises the hammer, but during this operation the water is discharged and the hammer falls on the rice placed below in a stone bowl.

The Formosans do not appear to make use of the method of raising water in buckets so common in the north of China. A small pool is made, into which the supply of water is led as close as possible to the irrigation canal, *i.e.* the lift is made as nearly vertical as possible. Two men stand at the high level, one on each side of the pool. They have a bucket generally made of close-plaited wicker work, which each holds by two strings, one attached to the rim and one to the bottom. The bucket is lowered to the water, the bottom strings being slightly pulled so as to cant the bucket and fill it with water. It is then lifted by the top strings, and just as it reaches the high level the bottom strings are pulled tight so as to turn the water out. This operation is performed with a quickness which is perfectly astounding.

About halfway between Tamsui and Banka, a river called the Kelung River runs into the Tamsui River. All along the upper portion of this river, as well as on the banks of the upper part of the Tamsui River, there are coal mines worked in a very primitive method by the Chinese; of these more will be said in speaking of Kelung.

A considerable quantity of coal is brought down to Tamsui for supplying the steamers which call at that port, but none is taken there for export.

A few miles up the Kelung River is situated the village of Sak-kow. In 1872 there was a stone there (and, though I did not see it, I believe it still exists), on which there is an inscription to the effect that coal mining is illegal. The change in the views of the mandarins since its erection is very remarkable.. The road between Banka and Kelung follows the valley of the Kelung River. This river has numerous rapids, but it is navigable for boats drawing 1 foot or 18 inches of water. Travellers going from Kelung to Banka generally go in boats, if possible, as it is much quieter and less fatiguing than going by road; but going in the other direction, much time is saved by adopting the road instead of the river. Coming down the river, the boat shoots the rapids; but going up, a pole is lashed across it, and three or four men wade in the stream and push against the pole. At about 20 miles from Banka the river passes within about 3 miles of Kelung, A ridge of hills lies between the river and the village. The road over is very steep, but the height of the summit of the pass is not more than 270 feet.

The last of the treaty ports to be described is Kelung. This is the most northerly of the open ports, and its harbour is the best in the island. It is, however, very small, there being room for only one or two vessels to swing. There is no bar, and there are 7 to 16 fathoms in the entrance, and 4 to 6 inside.

The Junk harbour is of considerable extent, and is close to the village. It is very shallow. It has been ruined by vessels discharging ballast into it. This practice is now forbidden, but it is not altogether stopped. The harbour is well sheltered from all winds except the north. The entrance is marked by an island, called Kelung Island, 580 feet high which forms a conspicuous object from the sea. There is deep water all round it. The harbour is very beautiful. Vegetation reaches to the water edge, and the hills are pretty wooded. There are numerous palms, and tree-ferns abound in the neighbourhood. There is a small island, called Palm Island, which shelters the harbour to a great extent. If a breakwater was built out from the western shore, opposite Palm Island, and some portions of the bay dredged, it would make the harbour a safe one in all weathers. This would cost money, but it would be a permanent improvement.

The rainfall at Kelung is large. In 1874 it amounted to 120 inches.

About 4 miles from Kelung there is a bay called Petaou Bay, or Coal Harbour. It derives the latter name from the fact of there being abundance of coal in the neighbourhood. It is a small but tolerably well-sheltered harbour. There is room for one or two small vessels, but there is no room for a large vessel to swing.

Between Kelung and Coal Harbour there are some very curious rocks, called the image rocks. These are comparatively soft rocks, with nodules of harder rocks imbedded in them. The sea has eaten away much of the soft rock, leaving the hard nodules projecting, and giving rise to most fantastic forms.

Coal has been worked in this district for a very long period by the Chinese. Their mines consist of little more than holes in the ground at the outcrop of the coal seam. The seam is followed as long as the mine can be kept dry. Arrangements are made for keeping the surface-water from running into the mine, and occasionally a wooden chain pump, like that used for irrigation, is employed to drain the mine, but it is practically correct to say that no pumping is employed, and wherever such is necessary the mine is abandoned. In one valley, near Kelung, there are said to be sixty such mines opened, and the number in the district is so great that 45,000 tons were shipped in foreign ships in 1873. This is the largest export as yet.

All vessels at present which come for coal lie in Kelung Harbour, the coal being carried to the water edge in baskets, and then shipped in cargo boats, which take it to the ship. Some of the coal has to be carried over the hills. Many children are employed in this work, which appears to be very hard.

Some time ago the provincial authorities determined to sink some pits and work the coal under foreign supervision. They engaged an English mining engineer, a Mr. Tyzac, who spent great part of 1875 in examining the district. Having determined on the proper course of procedure, he returned to England, and in May 1876, he despatched a party of eleven men, consisting of sinkers, miners, engineers, carpenters, and a blacksmith. All these were picked men, capable of superintending work in their several departments. A full supply of winding and pumping machinery, and everything necessary to start the work fairly, was sent out at the same time.

In April 1877, a shaft 250 feet deep, and 12 feet 7 inches in diameter, had been sunk, and was then within 50 feet of a bed of coal 3 feet thick, lying at an inclination of 1 in 6. No water to speak of had

been met with. The pit is being lined with sandstone of good quality, and all the permanent work and materials appear to be of the best description. The guides for the cage are of the new pattern, viz. vignoles rails to run at one side only. To all appearance there is more risk of a cage catching with guides of this pattern than with the old-fashioned wooden guides, but those supplied are good heavy rails, and, if well fitted up, will no doubt work well.

The mine is situated about 1½ mile from the coast, at an elevation of 150 feet above the sea level. A temporary tramway has been laid to the shore. The engines and machinery are capable of lifting 700 tons per day. Native workmen have been employed as sinkers at 30 cents per shift of eight hours, and have turned out well. The natives have also been trained to blast in the quarries without the presence of foreigners.

The Kelung coal is highly bituminous, and rather soft. It is a good steaming coal, in the sense that if employed there is no difficulty in keeping up steam, but it burns very fast, and a large quantity is required for a given amount of work. It is, therefore, unsuitable for steamers going long voyages. The smoke is very dense, and this is a drawback; and it is no uncommon thing to see flames rising several feet out of the funnel of a steamer. It is probable that the coal raised from the pit will be better than that mined at the outcrop, and there is little doubt that there will be a ready sale for all that can be raised. When it was determined to work the coal under foreign superintendence, a proclamation was issued saying that it was not intended to interfere with the working of the present mines, provided they did not encroach on the space set apart for scientific working.

Until lately the export duty on coal was 0·7 H.K. taels, say 4s. 6d. per ton; it has now been reduced to 0·07 H.K. taels, or one-tenth of the old amount. Including some local charges, the total dues payable on coal for export do not exceed 7d. per ton, an amount which brings some return to the revenue, and does not interfere with the trade.

Near Kelung, clay ironstone is found, but of very inferior quality.

There is a memorial-stone in Kelung erected to the memory of those who perished at the wreck of the 'Laptek.' This was a very distressing occurrence. The 'Laptek,' a British vessel, ran on shore about 8 miles from Kelung in 1874. All attempts to get her off having failed, she was surrendered to the underwriters, and Mr. Greig, Lloyd's agent, went to take charge. A lieutenant and several men from H.B.M.S. 'Kestrel' went on board to assist in dismantling. The work went smoothly for a long time, all on board having a pleasant time of it; indeed, it was described by one who assisted as being more like a prolonged picnic than like work. When the work was nearly finished, on 17th July, a severe storm rose suddenly, and Mr. Greig, the lieutenant of the 'Kestrel,' and six men were drowned.

In addition to the Treaty ports there are several native ports, but there is not much known about them. One of the best is Saw-o Bay, on the east coast. This is a tolerably well-sheltered port, with fair anchorage. The village is inhabited by Chinese fishermen. Twelve miles to the north of Taiwan-fu is situated the port of Kok-si-kou. This is a good port; but there is no inducement for foreign vessels to go there, even if the port was opened.

The port of Heong San, near Teuk Cham, is sheltered by sandbanks, but the water is shallow.

The Treaty ports of Formosa are points of such special interest that it appeared advisable to devote considerable space to a description of them; but there are several other cities and towns in various parts of the island which deserve notice.

The island may be divided, either according to its physical character or according to its inhabitants; but, practically, it makes little difference which method is adopted, because, roughly speaking, the flat portion is civilised, while the mountainous portion is savage.

Immediately to the north of Taiwan-fu, the country is sandy and slightly undulating. By degrees, as one goes north, the soil becomes richer; but for 20 miles the land is very poor, and badly cultivated. After this it improves quickly, and, before Kagee is reached, the land through which the road passes becomes rich and highly cultivated, producing rice, sugar, sweet potato, beans, betel-nuts, bananas, &c.

Very great attention has been paid to irrigation from this point right up to the north of the island, wherever the ground is at all suited to it. In many places the water has to be pumped, but in most places it has been brought from the higher ground in embanked canals; and wherever there are streams at a sufficiently high level, they have been trained, and have had weirs built across them so as to suit them for irrigation. Some of these weirs are considerable works.

Kagee, called Chin-lo-san by the natives, is a walled city of considerable importance. Paper is manufactured in the district, and the city is a centre of trade for the surrounding country. About 20 miles south of Kagee there is a small volcano, which has been seen by the missionaries. There is flame, but no smoke, issuing from the crater.

North of Kagee there are several rivers intersecting the rich alluvial plain. The first of any importance is the Boekhe, which runs about 20 miles north of Kagee. The bed of the river is composed of sand and gravel. It is very small in dry weather, but in floods it is a swift and wide torrent. It overflows the adjacent country sometimes to the extent of 1 mile in width.

A short distance to the north of this there are three rivers, the Chu-a-khe, the Twa-sin-tsung-khe, and the Potau-a-khe. The description of the Boekhe applies exactly to these rivers, except as to width. The flood bed of the first is 1½ mile wide; of the second, half-a-mile, and of the third, 1¾ mile. Included in these widths there are banks of sand and gravel, which may not be covered during floods, but they are merely composed of *débris* washed down by the stream, and are liable to be carried away at any moment.

After passing the Potau-a-khe, the road passes through very rich and carefully-irrigated rice lands, until Chang-hwa is reached. This is the second city in the island; but almost all the trade is carried on with the coast ports, which are not open to foreigners. Not only is there no trade between Chang-hwa and Taiwan-fu in foreign hands, but, as far as I could discover, there is no trade in native hands. Chang-hwa is a walled city, of apparently 60,000 or 80,000 in-

habitants; and judging from its busy appearance, the trade which it has with the coast ports must be very large. Very good bricks and tiles are made in the vicinity.

A few miles south of Chang-hwa there are signs of the ancient beds of mountain torrents, large quantities of gravel having been washed down into the plain. North of Chang-hwa there is more rice ground, through which runs the River Toa-to-khe, and one or two smaller streams. The hills to the north of this river approach much nearer the coast than the hills further south, and the road which to the south runs entirely in the plain, occasionally runs some little distance up the hills. The lower portion of the hills are being gradually brought into cultivation, and large quantities of rice are grown, but I saw no sugar in this part of the island.

About 20 miles north of Chang-hwa runs the Tyka-khe. This appears to be the swiftest of the Formosan mountain torrents. In the dry season it consists of a number of small streams, winding through banks of large shingle. At the time I crossed, all but one of these could be forded, but some of them were so swift that it was unsafe for one man to cross alone. Judging from the size of the stones, the river must be very fierce in floods, and the width of the bed liable to floods is 2 miles. Immediately to the north is the town of Tyka. Most of the trade of this place is carried on with the coast, but a little is carried on with Banka and the northern ports. Very fine grass mats are made here. A small trade is done in the second quality, but the very finest do not get into the market. They are about 6 feet by 4 feet, and are nearly waterproof. Immediately to the north runs the Pang-le-khe. This is a collection of mountain streams, all crossing from one gorge in the mountains. The beds of all are constantly changing. The width from the most southerly stream to the most northern is about 5 miles, and all the land between is liable to floods. There is a piece of land in the middle about half-a-mile wide which is cultivated, but it is composed of *débris* brought down by the streams, and though it has not been covered lately, it is liable to be eaten away by the rivers and re-deposited in some other place. The stones in the river are not so large as in the Tyka River, but the stream is very rapid in floods. And these rivers which have been described form great obstacles to the internal trade of the island. Travellers are often stopped for days, and, occasionally, I believe, for weeks.

The rainy season in the north of the island is in winter; and near Taiwan-fu, in summer; consequently, any one making the through journey is liable to be stopped somewhere. The best time for travelling seems to be October and November, and again in April.

North of Tyka, many of the villages are fishing villages. The hills come near the coast, and the cultivated land is broken up. Twenty miles to the north of Tyka is Oulan, a walled city, on the banks of a river, navigable by junks drawing little water. Near Oulan there is a desert of shifting sand, but it is of small extent. About 20 miles further north, Teuk Cham is reached. This is a walled city, of at least 50,000 inhabitants, and is the seat of Government of the Northern Provinces. Bullock carts can go from Teuk Cham to near Tamsui, and also to the native port of Heong San, a distance of 4 miles.

The sugar grown near Teuk Cham is considered to be good, but most of it must be exported in junks. Teuk Cham is also noted for its oil, made from the pea nut. This oil is largely used all over China, and burns brilliantly. It is by many considered superior to Colza.

North of Teuk Cham the road passes over some high downs. Much of the land is uncultivated, but, where water can be obtained, there rice is grown.

About 20 miles to the north of Teuk Cham lies Tien-lick, a large town, and a distributing centre for the surrounding country. There is a very large carrying trade between this city and Banka. About 10 miles north, the highest point on the road is reached, at a village called Gu-lu-na, after which there is a rapid descent into the valley of the Tokoham River, at a village called Sin Ching. This village is noted for its rice or pith paper. This paper is cut from the pith of the *Aralia papyrifera*. The pith is cut round and round, and the paper is wetted and pressed flat. The paper is exported principally to Canton.

About 3 miles below Sin Ching the Tokoham River joins the Tamsui River near Banka, the neighbourhood of which has been already described.

JAMES MORRISON.

*(To be continued.)*

## THE CROZET ISLANDS.

### (SOUTH INDIAN OCEAN.)

CAPTAIN LINDESAY BRINE, R.N., has made the following interesting Report on the Crozet Islands, in lat. 37° 9′ S., and lon. 150° 57′ E., which he visited when in command of H.M.S. 'Wolverene' in 1876:—

"We left Simons Bay on 18th November, and proceeded to the south until we fell in with the prevailing westerly winds in latitude 38° south, longitude 18° east. We then steered straight for the Crozet Islands until we reached latitude 45° 26′ south, longitude 38° 18′ east, the islands then being 622 miles distant. We now entered a dense fog, which continued, with rare intervals of open sky, until the morning of the 30th, at which time we had nearly run our estimated distance.

"We were on the point of heaving to, to wait for the weather to clear, when a sudden lift of mist disclosed for a few minutes the west end of the Twelve Apostles about three miles off on the bow.

"This point of the island is remarkable for its lofty group of pinnacle rocks closely resembling the Needles. These, although attached by their base to the island, appeared when first seen like sharp, jagged peaks, arising separately from the sea, and it was the low morning light penetrating between the open spaces that caused them to stand out so clearly. The island itself was not visible. A high rolling sea broke heavily on the weather shore.

"At about 9 of the forenoon the fog again lifted, and I steamed in for the land, to examine the Twelve Apostles (the island on which the 'Strathmore' was wrecked), firing guns to attract attention.

"It was evident that no close observation could be made unless a change took place in the weather; but most fortunately the fog cleared away, and was succeeded by two singularly fine and calm days, and

ceeded in this, they abandoned the design and removed to St. Mary's Island on the east side of Madagascar.

In 1799, during the war with the French in Egypt, Perim was occupied by a British force, in order to prevent the French coming down the Red Sea and attacking our Indian possessions. This force, called the Red Sea detachment, consisted in all of about 300 European and native troops, and was commanded by Lieutenant-Colonel J. Murray, R.E., who was appointed Political Commissioner for the Red Sea. He arrived in Perim Harbour on the 3rd May 1799, and on the same evening took formal possession of the island in the name of the East India Company. A few days afterwards a naval force, under the command of Commodore Blankett, arrived from Great Britain with orders to assist Colonel Murray in holding the entrance to the Red Sea. The detachment had not been long on the island before the want of water began to be felt, and though every effort was made to procure water on the island, it proved unsuccessful, and on the 8th September Colonel Murray withdrew the greater portion of his troops to Aden. Numerous traces of Murray's occupation still remain on the island, including a ruined circular tank, intended to hold 25,000 gallons.

During the year 1835 a good deal of correspondence passed between General Coghlan, then Political Resident at Aden, and the Government of Bombay relative to the erection of a lighthouse at Perim, and in the following year Sir Henry Locke, Commander-in-Chief of the Indian Navy, strongly recommended that this step should be taken, and that Perim should be occupied by us for the purpose. In the course of his letter he pointed out that this island was the key to the Red Sea, and that in the event of a war it might form a place of resort for privateers. "Again (he continued), in the event of a ship canal being cut between Suez and Alexandria, the value of such an island to the interests of trade is very apparent and of the utmost importance to England and India." Sir Henry added a significant postcript to the end of his letter, stating that he had just heard that a French engineer* was about to survey Perim and the islands in the Red Sea. This intelligence induced the Government to pass a resolution in the Secret Department on the 10th November 1838, directing the immediate occupation of Perim. Lieutenant Greig, R.E., Assistant-Engineer at Aden, with a detachment of Sappers and Miners left Aden for Perim in the H.E.I.C. schooner 'Maki' on the night of the 13th January 1857, and on arriving there built a defensible barrack or block-house. He also made an accurate plan on the scale of 500 feet to an inch, but a fresh one, with contours 12½ feet apart is much needed. Lieutenant King, from whose description our article is borrowed, spent eleven months on the island and fully explored every part of it.

* This was M. Lambert, agent of the firm of Messrs. Lambert and Co., of Bombay and Mauritius. He had during the past twelve months visited and surveyed Perim and did not hesitate to state that he had recommended the French Government to occupy it.

## A DESCRIPTION OF THE ISLAND OF FORMOSA.

### WITH SOME REMARKS ON ITS PAST HISTORY, ITS PRESENT CONDITION, AND ITS FUTURE PROSPECTS.

*(Continued from page 266 of our last number).*

THE one common characteristic of all the villages (a characteristic from which, unfortunately, the cities are not free) is dirtiness. It is hard to say whether the inside or outside of the houses is worst. As in other parts of China, the night soil is collected in pits, public latrines being erected at various places. These, however, are never cleared out, the tubs for the fields being filled as required, and the whole neighbourhood of one of these pestilential holes is a mass of festering putrefaction. Tubs are also placed in the streets and courtyards, with very little regard to decency.

At most of the villages there are one or two inns. The sleeping rooms are generally 10 or 12 feet square, and contain two beds. The beds are of bamboo, and are fixtures. Fortunately, the only bedding is a mat. Sometimes there is a hole through the wall—more than half filled with wooden bars—which does duty as a window; but the rooms are invariably so dark that a lamp is required even during the day. The lamp consists of a small cup of oil, with the pith of a rush projecting a short distance over the edge. The lamp gives a great deal of smell, and enough light to smoke by.

The inhabitants have no idea of the advantage of fresh air. On one occasion, in company with Mr. Elles, I entered one of these rooms in Ung Kang Bey. The entrance was through a cooking and eating room, into a court partially roofed over, and filled with refuse and filth of the most disgusting description. The door of the bedroom opened into this court, and the window consisted of one tile replaced by a few pieces of oyster-shell. The room being oppressively close, I tried to explain to the proprietor that I would like some air, and suggested removing some tiles. After some time, an idea seemed to strike him, and, mounting on a stool, he pulled aside one tile that was made to slide. From his delighted air, and from the way in which he kept pointing at it, and ejaculating "Haw! haw!" I came to the conclusion that this was a unique contrivance, and was probably used for the first time. The size of the opening was 5 inches by 3.

The want of a good road ought to be severely felt, but I doubt if many of the inhabitants really think that anything better than the present road is required. The main road from north to south varies from 3 to 8 feet in width, where it is visible: in some places it is lost in a maze of paths. In the northern part of the island it is paved with round water-worn stones, driven in so as to form a causeway; and near Kagee it is paved with flat stones, about a couple of feet wide. In many places, however, it is merely a mud track, almost impassable in bad weather. Near Taiwan-fu, and also near Teukcham, bullock carts are used. As a rule, these can be used, not in the districts where the roads are good, but in the districts where the land is poor and there is little cultivation, and, consequently, the carts can leave the track in bad weather. There is no road on the island suffi-

ciently good to sustain regular cart traffic at all seasons.

The country which has just been described is the flat and civilised portion of the island. The hilly and uncivilised portion is of quite a different character. The west coast of the island is so low that it can hardly be seen at a distance of 3 or 4 miles, and houses and trees near the coast are liable to be mistaken for junks at a very slight distance. The east coast, on the other hand, is generally mountainous, with deep water very close to the shore. A ridge of mountains runs up the centre of the island. Mount Morrison, the highest of the range, is said to be 12,850 feet, and Mount Sylvia 11,300 feet. Mount Morrison is not named after the well-known missionary and Chinese scholar, but after the captain of one of the early vessels trading to Taiwan-fu.

All the land east of the mountains is supposed to be hilly, and inhabited by savages.

About 20 miles to the east of Chang-hwa there is a valley called Posia. This valley contains thirty-two villages, inhabited by semi-civilised savages. No foreigner resides there, but there is a branch of the Presbyterian mission, and the place is regularly visited by the foreign missionaries. Besides them, a few foreigners have penetrated there from time to time. Until lately the road to Posia was very bad, but it has recently been much improved. From Ta-niau, a village 6 miles north of Kagee, a tolerable path leads eastward to Chip-chip. A road from this point to Posia was commenced by the Chinese in 1876, and completed early in 1877. This road runs along the side of the hills, and is made by cutting a narrow path out of the slope, and using the material to increase the width by embankment. Before the road was finished, the southern portion was much injured by rains, the embankment in many places being entirely washed away. The road passes near a lake, which is about 3000 feet above sea level. The borders of the lake are inhabited by a small tribe called Shwui-hwans, or water savages. The hills round Posia are carefully cultivated in terraces. Millet is largely grown, also a fine quality of hemp used for making grass cloth.

For some time the Chinese have had military stationed on the borders of the savage country, trying to open up communication with the east coast. Sau-o Bay, on the east coast, and all to the north of it, have been in Chinese hands for many years. South of it, there are two Chinese settlements, called Sin-cheng and Khe-lai. The most northerly has been in existence twenty years, but has never numbered more than a hundred Chinese. In Khe-lai, on the other hand, there are about 700 Chinese and about 1000 semi-civilised savages.

In 1875, the military constructed a road running along the coast for about 60 miles south from Sau-o Bay, and passing through these settlements. Early in 1877 it was reported that the military from Chip-chip had constructed a road to meet them, and had effected a junction.

The people who inhabit the two parts of the island differ as much in their characteristics as do the two portions of country. The inhabitants of all the level country are Chinese. They have come from all parts of China, but principally from the Fokien province. The colloquial language closely resembles, or, indeed, is almost identical with that of Amoy. The separate clans of Chinese were at one time almost as hostile to each other as to the aborigines, but they are now on much better terms, and to the foreigner the Chinese appear to be divided into two classes only, viz., the ordinary Chinese and the Hakkas. The Hakkas emigrated from the province of Kwang-tung some years ago. A great many are to be found inhabiting the villages in the south of the island. Ten or fifteen years ago these people lived entirely by plunder, and at the time the ports were first opened it was as much as a foreigner's life was worth to pass a night in one of their villages. They are a hardy race, and if they do not settle down to quiet industrial occupations so readily as the other Chinese, they have more daring, and are much more enterprising. In many parts of the island they are to be found acting as go-betweens with the savages. In going north by the main road from Taiwan-fu, they are first met with about Tong-siau, some 12 miles north of Tyka. From this point they are met with at intervals right up to the north of the island. There is a large settlement of them at Tokoham, where they are engaged in the preparation of camphor. There is a great contrast between the Hakka women and their Chinese sisters. The Hakkas never contract the feet, and consequently can walk well. As a result they have sometimes to do hard work in the way of carrying heavy bundles which no other Chinese woman could lift, or at least walk with; but in spite of this they have always a cheerful expression. They are well made, and in many cases handsome. They wear loose trousers, and walk with bare feet. The Chinese women are not unkindly treated, and the compression of the feet does not seem to have any evil effect, but it tends to make them helpless and unable to take proper exercise, and consequently they have often a sickly appearance.

In culture and general intelligence, the Chinese people of Formosa seem to be on a par with those on the mainland. The fact that wages are 20 per cent. higher than in Amoy is a proof that there is no lack of money in the island, though the fact of opium representing two-thirds of the value of all their foreign imports show that they do not make the best possible use of their wealth.

In speaking of distances they use the words *li* and *pau*, the latter being roughly about 10 *li*. As in other parts of China where foreigners have settled, a *li* is generally understood to mean one-third of an English mile. I believe, however, it corresponds better to such an expression as "a five minutes' walk," and did not originally represent a certain length. Mr. Baber says that in Szechuen he found places where the distance from A to B was 20 *li*, but from B to A 40 *li*. On questioning some people he was told that as it was up hill in one direction it would not be fair to the coolies if the length were stated the same in both directions. He suggested that the coolies sometimes had to work by night or in bad weather, but was met by the answer, "O! that is arranged for in the price." The best definition I can give of a *li* is the distance a coolie will walk with a burden in six minutes. A *pau* appears to mean the distance the chair coolies carry without setting down the chair. It is about fifty minutes or an hour's walk. Europeans are, how-

ever, impressing on the people the necessity of having clear measures of distance.

The laxity which prevails in civilised countries regarding weights has had its effect in inducing foreigners to submit quietly to the Chinese practices in this particular. A *picul* of camphor is expected to weigh far more than 100 *catties*. Sometimes as many as 150 are demanded. This is supposed to be an allowance for the loss in pressing or on the voyage. If asked how difference of quality was allowed for, the merchant would no doubt complacently answer, "O! that is arranged for in the price." The standard *picul*, by which every thing should be weighed, consists of 100 *catties* of $1\frac{1}{3}$ lb. avoirdupois, or $133\frac{1}{3}$ lbs.

In the matter of sawing wood they follow the plan of sawing on a horizontal plane; at least, I saw this system followed at Taiwan-fu, and I did not notice any other system in use in any other part of the island. This matter is worthy of notice. All over China the usual saw is a framed saw. The teeth are formed, not, as in our saws, to cut all in one direction, but one half of the teeth cut in one direction and the other half in the other direction, the change being made in the middle. The half of the saw farthest from the sawyer cuts when he is pushing, and the half nearest when he is pulling. In different parts of China the saw is used in different ways. In Shanghai, when a log has to be sawn, one end is raised from the ground and a trestle is placed under it near the middle, the lower end of the log being sometimes weighted. The sawing is commenced at the raised end, the saw being used as in the ordinary English saw pit, with this exception, that instead of being placed in the centre of a frame it is placed at one side, like the small saws used at home by cabinet-makers. When the saw draught has reached the middle of the log, the other end is raised, and a draught is carried to meet the first. I have had logs nearly 70 feet long sawn in this way with a saw not more than 3 inches wide. The work is, as one might suppose, indifferently done, the saw draught being always more or less twisted.

In Chefoo the same description of saw is used, but the log is laid on two trestles, and the saw draught is in a *horizontal* plane, the men standing or squatting one on each side. The log, when small, has to be fastened by ropes or by heavy weights laid on it.

In Tientsin, a large quantity of timber is imported in round logs 10 or 12 feet long and 4 feet or more in diameter. These logs are set up on end, the men mount on stages and saw in a vertical plane from top to bottom. When the log is too small or too light to stand in this way it is fastened up, sometimes not without difficulty. This method is capable of giving fair results, and is used for small pieces of wood in the shops in Shanghai, Canton, and other places. The sawing of large logs by this method is, as far as my observation has gone, confined to Tientsin. In Pekin I have seen the Chefoo method used, and, as above remarked, I have seen it in Formosa. It has one peculiarity which may recommend it to a republican mind—both men are equal, there is no "top sawyer."

Opium-smoking prevails to an extraordinary extent in Formosa; but, as far as one could judge from a short visit, there is not a great amount of smoking to excess. In travelling through the country, my chair coolies always smoked opium at night, and continued smoking after I had gone to bed; yet they were always ready to start before six o'clock in the morning, and seemed fresh. Any such coolie will carry 20 miles a day for ten consecutive days. Smoking in this way costs from 10 to 15 cents per day, and, when a man's wages are only 20 cents a day, opium-smoking soon means ruin; but the chair coolies get at least 75 cents when employed all day carrying, and, therefore, can afford it.

Of the savage inhabitants very little is known. Shipwrecked sailors who have lived among them have given some account of their life in captivity, but this is the only available source of information. Of the semi-civilised savages, however, we have accounts from missionaries and others who have visited them. These semi-civilised tribes go by the name of Pepohwans in the south of the island, Sekhwans in the centre, and Paktowhwans in the north; but these distinctions are seldom adhered to, and all over Formosa they are known by the name of Pepohwans. The Chinese appoint a head man for each village or district inhabited by these people, and he settles disputes according to his judgment; but there appears to be no law. Of late years the savages have been tolerably peaceable, and have carried on a considerable bartering trade with the Chinese; but, quite recently, an edict has been issued forbidding the sale to the savages of guns, powder, or lead. The sale is still going on surreptitiously; but, if it is entirely stopped, there is likely to be some trouble.

The savage women do some handsome embroidery. They are very fond of pieces of coloured cloth, which they pull to pieces, using the threads for this work. They also unravel coloured cloth—by preference, red, green, and orange—and weave or plait the threads into tri-coloured bands for turbans.

Formosa is the seat of two Presbyterian Missions, one connected with the English Presbyterian Church, and the other with the Canadian Church. The English Mission has its head-quarters at Taiwan-fu. The Rev. Dr. Dickson and the Rev. W. Campbell are stationed there. Both of these gentlemen have travelled over much of the island where no other foreigners have penetrated. Mr. Campbell was the first, or one of the first, to enter Posia. Dr. Dickson is a medical missionary, and he has established hospitals at Takao and Taiwan-fu, which are much appreciated by the Chinese. He is often asked by soldiers to prescribe for them, and has offered to attend a military hospital if the authorities will build one, but nothing has yet been done. The Canadian Mission is under the charge of Dr. Mackay. This gentleman lives near Banka, in the north of the island, and is on extremely good terms with all the Chinese authorities there.

The usual mode of travelling in Formosa is by chair. Ponies may be used for riding short distances, but the numerous rivers, too deep to ford and too rapid to swim, render them useless for long journeys, many of the frail rafts or boats which are not very unsafe with one or two chairs being quite unsuited for ponies. The Formosan chair is very light, and may be comfortable for a Chinaman. It is 40 inches long, 48 inches high in the centre, and 40 inches at the sides, and 21 inches wide inside, with a seat about 10 inches high. These dimensions are enough to

show that there is not much room for an ordinary-sized Englishman to move about; still, with the aid of a thick Chinese blanket, the chair can be made tolerably comfortable. The method of carrrying is simply diabolical. Four men carry, two being placed at the ends of the poles, and two close to the chair, one in front and one behind, the two latter supporting the chair by means of cross pieces or yokes passing over the shoulders and attached to the poles by ropes. These ropes are so adjusted that when the men stand still and the chair is loaded, the poles are bent about 3 inches. In carrying, the easy swing of the ordinary chair is lost, for just as the Formosan chair is coming down gently, it is brought up sharp by the men close to it. The motion is exactly the same as is used for jigging crushed metallic ores.

The chairs in Canton are carried in somewhat the same manner, but the poles are better, and the motion is not nearly so disagreeable. The proper method of carrying by four men, with balance bars, as used in Hongkong and in many parts of China, is unknown in Formosa, except in the case of the chairs of a few customs officials. Chinese are sometimes carried by two men, but no foreigner can induce the natives to carry him in that way, even if he offers to provide relays. For short distances three men are employed. This is somewhat worse than four. It is true that, after some practice, one gets accustomed to the motion, and can read and, I believe, sleep; but at first it is very trying.

Foreigners, as a rule, walk a great deal—say, half the day or more—when travelling; but it is necessary, or, at all events, convenient, to have a chair for crossing the shallow rivers. Twenty miles is a fair day's journey, if the men carry all day; but 25, or, at a push, 30 miles may be done, if the traveller walks much. It is only in rare cases, however, that a foreign traveller wishes to go so fast.

In travelling through the civilised part of the island, it is a good thing to take a proper supply of provisions. Carrying coolies can easily be obtained, and no advantage is to be derived from going badly provided, except the advantage of being more than usually uncomfortable. Coolies can be engaged at a cost to the traveller of about $1 for each per day. Having decided how many coolies will be required, the best plan is to make a bargain with a head man for the whole journey. He will then go through the whole way, changing coolies, if necessary, at the villages on the road; but this does not affect the contract. Fowls, eggs, and sometimes fish can be obtained on the road; but bread, tinned meat, cheese, spirits, &c., must be carried. Two coolies can easily carry provisions for two travellers for a week or ten days.

It is true that a person who has been long resident in the country, and who knows the language, may see more if he travels alone, and may learn something of the ordinary home life of the Chinese, from which the traveller with his string of coolies would be entirely shut out, and he will, moreover, effect a considerable saving in expense; but the ordinary foreign traveller would, probably, see less if unattended than if he had the usual retinue, and the saving of expense would be small compared with the total cost of a journey to the island. A boy acquainted with the language should accompany the party, and should have a chair. He will then be able to cook and serve up meals. It must, of course, be understood that these remarks refer only to the civilised portion of the island. When the traveller makes up his mind to enter the borders of the savage country, he must cut down his baggage to the smallest possible dimensions, and, if he carries provisions, should only carry a small supply to guard against emergencies, and not for general use. And here it may be well to speak of carrying a revolver or other weapon. In a country where a traveller is likely to be attacked by brigands or robbers, a stout heart and a good revolver may carry him safely through many difficulties; and in the savage part of Formosa, as in all savage countries, a certain amount of respect will be given to any one who goes about evidently prepared to fight for his rights. In the civilised part of Formosa, however, or in any part of China where there is a fair amount of civilisation, a revolver is worse than useless. If any serious difficulty arises, it will be with the population in general, or with large numbers of them, and against such odds fire-arms must be not only useless, but dangerous. In the first place, if used rashly, it may cause a petty squabble to assume serious proportions, in which case lives may be lost, that of the traveller included. In the second place, people are more ready to resort to violent measures with a man who carries a weapon that at any moment may be used to wound or kill, than with a man who is wholly unarmed. I have carried a revolver for weeks on the borders of Greece, sleeping with it under my pillow at night; and I have travelled without one in many places, including Formosa (though I took one to the island), and the more I see and hear of travelling in thickly-populated countries, the more I think arms of any sort unnecessary encumbrances.

I am by no means of those who say that in all cases of a squabble between foreigners and Chinese the former are to blame; still a foreigner may often take offence where none is intended, and so unwittingly give rise to disputes. The extreme curiosity of the Chinese is very trying. They crowd round one, and take special notice of every movement. One cannot blow one's nose, or make an entry in one's note-book, without giving a subject of conversation to scores of spectators. At meals they will crowd round one out of doors until one can hardly breathe, and indoors they will crowd into the room, or, if it be possible to keep them out of that, they will apply their eyes to every crack and crevice, of which there are sure to be plenty. No one who has not experienced the annoyance of this knows how trying it is. The first time the novelty of the scene may somewhat amuse, and possibly, after years of experience, one may become hardened. On this latter point, I can only say eighteen months has not been long enough to harden me. I know by experience that it is possible to control one's temper, so as to avoid striking; but if any traveller were to strike a native under such circumstances, I should be far from saying there were no extenuating circumstances; and yet, if trouble ensued, it would be necessary to treat the case as one in which the foreigner was in fault, for a hundred chances to one the native had not the slightest idea he was even being rude.

JAMES MORRISON.

*(To be continued.)*

broad billows. The line of rapids occupies a distance of 180 miles, thundering down steep after steep, in a succession of 62 cataracts. Mr. Stanley gives the fall at 585 feet. It took the party five months to pass these cataracts. Here the gallant young Englishman, Frank Pocock, lost his life, as well as fifteen of the men, and Stanley then hauled his boat and canoes on the shore, above the cataract of Isangila. From the rapids the discoveries of Captain Tuckey, in 1816, commence.

We regret that Mr. Stanley should have gone out of his way to assail the memory of Captain Tuckey, especially as his statements respecting Tuckey's map are incorrect. He says that Captain Tuckey "placed on his map, east of Yellala, about E. long., 16° a cataract at a very wide interval from Yellala Falls, with names of places overland as if he had quitted Yellala Falls and proceeded overland as far as the Kwango river." Captain Tuckey did nothing of the kind. The most easterly rapids on his map are west of Covinda, in about 14° 45' E. longitude; he expressly says that he never got further east than Soundy N'Sanga in about 15° 10' E.; and he puts no names of places overland to the Kwango river. There is nothing in Captain Tuckey's map and narrative to mislead any one; and the only way to account for Mr. Stanley's extraordinary statement is that he was provided with some incorrect manuscript compilation, and that he had never seen Tuckey's map. Captain Tuckey was a good surveyor and a most accurate and careful observer. The imputations against him are entirely unfounded; and it is the duty of geographers to defend from unjust aspersion the memory of a gallant explorer who died, from exhaustion and illness, in the faithful discharge of a perilous service.

Sir Roderick Murchison, whose forecasts were seldom wrong, inclined to the opinion that the Lualaba of Livingstone was the Congo. We fully discussed the question, arriving at the same conclusion, in our number for September 1872, p. 174. Subsequently, Dr. Behm advocated the same view. Mr. Stanley by descending the river, has established the fact beyond further dispute. The distance from Nyangwe to the mouth of the Congo, following the course of the river, is calculated at 1800 miles. The highest rise of the river was from the 8th to the 22nd of May. The depth of the rise in the broad parts was about 12 feet, and in the narrows from 20 to 30 feet. Ivory is described as being very abundant, and there are extensive groves of oil palms.

Mr. Stanley was navigating the Congo for nine months, from the 5th of November 1876 to the 11th of August 1877. The difficulties which were encountered and overcome were extraordinary. Dangers in every form were met with intrepid resolution. Marvellous resource and ingenuity were exercised in combating the great physical obstacles, still more in resisting the incessant attacks of the savages. The achievement thus done was of the highest geographical importance, and places Mr. Stanley in the first rank among African discoverers. Captain Purvis, of H.M.S. 'Danae,' placed H.M.S. 'Industry' at the explorer's disposal when he arrived at Loanda, and he and his African followers arrived in her at Simons Bay on the 22nd of last October, proceeding thence to Zanzibar, on the 6th of November.

## A DESCRIPTION OF THE ISLAND OF FORMOSA.

### WITH SOME REMARKS ON ITS PAST HISTORY, ITS PRESENT CONDITION, AND ITS FUTURE PROSPECTS.

*(Continued from page 296 of our last number.)*

THE present condition and future prospects of Formosa furnish a topic of great interest. From early days the stories of its riches have attracted many emigrants, and the flat portion of the country is now thickly populated. The cultivation of rice may be said to be perfect. The systems of irrigation are not the result of individual exertions, where each man works for himself, but are clearly the work of large bodies of the inhabitants working for the common good, probably at the instigation of the local Government, and certainly under the direction of good masters. Under the stimulating effect of foreign connections, the trade of the island, by no means despicable to begin with, has been more than doubled in ten years. Not only have old industries flourished, but new ones—pre-eminently tea-cultivation—have been created, and the general state of the Chinese portion of the island may be described as prosperous. Some details as to the trade of the last few years will be interesting.

According to the consular reports for 1874, the total trade of Formosa in foreign vessels was:—

| | 1873. | 1874. |
|---|---|---|
| Imports ... ... ... | £602,012 | £680,340 |
| Exports ... ... ... | 485,259 | 594,431 |
| Total ... ... | £1,087,271 | £1,274,771 |

The total value of later years is not yet published. By far the most valuable of the imports is opium, two-thirds of the total value being due to this drug. Next in importance come cotton piece goods and woollens. The total number of pieces of all descriptions imported was 88,523 in 1873, and in 1876 the number had risen to 146,507.

Among the exports, tea, camphor, and coal at Tamsui and Kelung, and sugar at Taiwan-fu or Takow, account for almost the total value. The quantities, as stated in the *Customs' Gazette*, are as follows:—

| | 1873. | 1874. | 1875. | 1876. |
|---|---|---|---|---|
| Tea, piculs*... ... | 15,610 | 24,610 | 41,573 | 58,905 |
| Camphor, piculs ... | 10,755 | 12,080 | 7,139 | 8,794 |
| Coal, tons ... ... | 45,177 | 15,221 | 27,665 | 31,593 |
| Sugar, piculs ... | 402,857 | 686,220 | 487,084 | 879,943 |

In addition to the above, some small quantities of tea were shipped at Taiwan-fu, and of sugar at Tamsui; but the amounts were quite insignificant.

There are no statistics of the trade in native vessels; but the foreign Commissioner of Customs, in his report for the year 1875, states that during the year 15,000 tons of coal were exported from Kelung in junks, and that there were exported by similar means from Tamsui 210,000 tubs of indigo, value $2,520,000, and 160,000 piculs of rice, value $320,000, besides minor exports. The amount of rice exported from the coast is enormous. So far back as 1834, it was estimated that 200 junks were employed in this trade. I do not know of any reliable estimate of recent date,

* A picul is 133⅓ lbs. avoirdupois.

but merchants talk of the large number of junks so employed. In 1872 the crop of rice failed, and rice had actually to be imported. It is not likely that this will occur again, as year by year the cultivation, already good, is becoming better and more extended.

The statistics of trade given above are most encouraging, and it is certain that the limit of prosperity has not been reached. There is still room for much improvement, and there is every reason to expect that the next ten years will see as great an advance as the last ten. The island has, ever since its acquisition, been considered part of the Fokien Province. The Governor-General who rules over that province ought to visit the island once every three years. His visits, however, have been irregular.

Ting, lately Futai of Foochow, has been appointed Governor of Formosa, and it is said that he is anxious to have the island made into a separate province, and himself appointed first Governor-General. Ting is a most enlightened Chinaman. He has a considerable knowledge of foreign manners and customs, and is ready and willing to introduce foreign improvements. He has a little of the jealousy of foreigners which is universal among the higher class of Chinese; but he has much less of it than Li Hung Chang, and most other high officials. It is a little difficult to describe the exact position of these men with regard to progress. Perhaps the nearest approach to a correct description would be to say that they are pro-foreign and anti-foreigner. To any one coming from Europe, accustomed to hear of Li Hung Chang as the champion of progress in China, it sounds strange and inexplicable to hear residents in China talk of his anti-foreign tendencies. Men of his stamp would like to have steamers and railways, but detest foreign steamboat or railway companies, and, if they had the choice of steamers and foreigners, or no steamers and no foreigners, would not hesitate a moment to choose the latter. Progress they are in favour of, but China unassisted cannot progress quickly, and slow progress without foreign assistance commends itself much more strongly to these people, than quick progress where foreigners are the moving spirits. Under Ting as Governor, or Governor-General, there is, therefore, more chance of a province progressing quickly than under most men; but there is another reason to hope for quick progress in the case of Formosa.

Although a governor-general of a province has a great deal of power, he is not absolute ruler, and has to apply to Pekin for permission to make radical changes in his province, or to carry out any important work, and further, he has to report all that occurs to Pekin; and even in those cases where special permission may not be required, the Imperial Government may consider that the provincial governor has acted imprudently, and may remove him and substitute another who will rule more in accordance with their views.

There seems, however, to be an understanding that Formosa is to be managed differently from the mainland. It is separated from China proper by 120 miles of sea, and it is thought that it is a suitable place for experiments. If these experiments lead to any bad result, it will be annoying, but will not be ruinous. The Customs' reports for 1875 contain a most interesting and able account by Mr. Brown, of Takow, on the advances that are being made in the way of constructing roads and improving the administration of the southern provinces, and the report of Mr. Hobson deals fully with the origin and progress of the mining scheme referred to in the earlier part of this paper.

Ting, since his appointment, has been making inquiries as to the cost of railways and telegraphs. The time has hardly come when works of that description in Formosa can be expected to give a direct return commensurate with the outlay, but if the Government can spare a little money now and again to construct roads, bridges, and, if possible, railways and telegraphs, the effect will be immediately seen in the Customs' returns, which are the best indications of the prosperity of the island.

Since the above was written a memorial from Ting has appeared in the *Pekin Gazette* (translated in Shanghai 13th June 1877), asking permission to make alterations in the assessment of various districts of the island for two reasons: first, the inroads of the sea, floods, and other natural causes, have in many cases considerably altered the area and value of the districts; and, secondly, the increased value of recently-established prefectures has made the weight of taxation bear most unequally on the various inhabitants. Certain new prefectures are double the area, and pay half the tax, of certain older ones, though now both new and old are equally well and fully cultivated. He also asks permission absolutely to abolish a certain class of petty imposts of very old standing, which bring little to the revenue, and press heavily on the people, more particularly as additional amounts are extorted from the people under cover of these taxes by the local mandarins and middlemen. He observes that the increasing customs and lekin duties may be counted on to recoup the treasury for any loss they may incur by these changes. The testimony of all who know China well, whether they belong to the commercial or official class, is so unanimous to the effect that stubbornness, stupidity, and conceit are among the most marked characteristics of the Chinese, that there is no escape from the conclusion that their views are correct—still, if even one man can be found who looks for an increase of revenue through abolishing oppressive taxation, and if the Government show their appreciation of him by appointing him to a high office, the state of China cannot be absolutely hopeless.

Two or three of the present industries offer splendid fields for foreign enterprise, if reasonable facilities are granted. The cultivation of tea might be extended to an unlimited extent, and the example of India leads one to suppose—I might almost say proves to one—that the value of the leaf might be increased twofold. The cultivation has been fostered by foreign intercourse, and foreigners take a more active part in its preparation for the market than they do on the mainland of China; but the tendency at present, unfortunately, is to assimilate the practice to that of the mainland, and not to that of India. The tea in Formosa is grown in very small lots, and the growers have not sufficient capital to cultivate it so as to give the best results, even if they had the knowledge of how it should be done.

Indigo also appears to offer an opening for foreign enterprise. The 210,000 tubs exported from Tamsui in 1875 were valued at $2,520,000, or $12 per tub.

Estimating the weight of a tub at 3 cwt. this would give $4 per cwt. The indigo imported into England must be worth about 30*l.* per cwt. In the consular report for Tamsui for 1871 an analysis of Formosan indigo is given, copied from the *Chinese Recorder* for July 1870. It is as follows:—Water, 5; sand, 1; carbonate of lime, 1; indigo, 1. Assuming the colouring matter of the Formosan indigo to form one-eighth of the whole weight, and the colouring matter of that imported into England to form half of the weight, these prices give $32, or 6*l.* 8s., per cwt. as the price of the colouring matter in one case, and 60*l.* in the other. Surely there is a sufficient margin here for profit. Formosan indigo in its present state is quite unsaleable in England; but judging from the fact that it is found remunerative to print goods in England in a way to resemble indigo printing, it is probable that if Formosan indigo were improved, it would find a market, even if it were not as good as Indian indigo.

Sugar may also offer some slight inducement to foreigners. As far as I could judge from all I heard and saw, there is not much room for improvement in the growing of the cane. The preparation of the sugar is, however, imperfect in the extreme. The mills are very primitive. The cane has to be passed twice through the rollers, and even then all the juice is not extracted. The natives have got into the habit of adding much more lime than is necessary in the boiling, and the resulting sugar contains a large quantity of it, and soon destroys the charcoal in the refineries. From the moment, then, that the cane is cut, the whole process is open to great improvement. Some attempt has been made to introduce sugar-crushing machinery, and there are two Glasgow-made crushing-machines on the island. They are, however, very small, and actually in some respects inferior to the ordinary mills of the country: they consequently have not been favourably looked upon. If powerful steam-crushing machines were introduced, they would effect a great change, and, under foreign supervision, the crystallised sugar would contain much less dirt than at present. On the whole, however, sugar preparation offers a better field to the native than to the foreigner; and at least one really wealthy Chinaman has offered to buy a good steam sugar-crushing machine, if it can be shown to him at work. As this man has lately paid $500,000—or, say, 100,000*l.*—towards a fund for making roads, &c., without apparently losing his reputation of being the richest man in the island, he is well able to undertake the introduction of improvements on a large scale if he likes.

Of new industries, the most hopeful appears to be coffee. There can be no reasonable doubt that on the southern hills it might be cultivated to a profit; but it is almost a certainty that nothing can be done, unless sufficient facilities are granted to induce the investment of foreign capital.

I must now refer to minerals. The mineral wealth of Formosa has always been spoken of as immense, and, before visiting it, I had seen specimens of iron-stone and other minerals which had undoubtedly been found on the island. It is, therefore, certain that some useful minerals do exist there, and it is not improbable that in the savage country there may be large and valuable mineral districts. I believe, however, I am correct in saying that absolutely nothing is known of them. There is a story current among the Chinese that there is a jade mountain in Formosa. The existence of this mountain was never believed in by foreigners, but a report came to my ears through a foreign channel that mercury was to be found on the island. I mentioned this to the mandarin who has lately been sent to take charge of the sulphur mines, and asked him if he knew anything of it. He is a man of considerable attainments and high position, and his knowledge of the English language and long connection with foreigners have given him opportunities of learning much that most Chinamen are ignorant of; nevertheless his answer was—"Yes; it is seen to glitter on the mountains." Such an answer not only proved that he was utterly ignorant of the whole subject, but showed clearly that one must be careful in accepting vague statements on such matters, even if they come from apparently good sources.

Some valuable minerals are by no means attractive in appearance; but galena, copper pyrites, gold, and many others have such distinctive characters, that specimens are sure to be picked up by the natives of any country where they are to be found; while the weight of ironstone of all descriptions, of tinstone, and of other metallic ores is likely to attract attention. I took every opportunity of inquiring whether any of these were known to exist; but no one resident in the island, either Chinese or foreigner, could produce a single specimen, with the exception of Mr. Tyzack, who kindly supplied me with a nodule of clay iron-stone. It was of poor quality, but he had been able to find nothing better.

Coal, sulphur, and petroleum are the only mineral productions whose existence in considerable quantities has been proved. The coal will probably continue to be worked by the Government, and the sulphur, as I have already stated, will certainly (and I think quite properly) be kept in their own hands for the supply of their powder-mills. Petroleum exists about 10 miles south-east of Oulan. Some years ago the right of sinking wells was granted to some foreigners, and the petroleum obtained was stated to be worth 15*l.* per ton; but this grant appears to have been withdrawn. Of the rights of this case I know nothing, but I was told by some of the high officials in the island that they intended to work the wells themselves, and since leaving I hear a report that they have sent for an experienced man from America to take charge. At the present moment, then, minerals do not offer as great inducements to foreigners as tea, coffee, or indigo.

If, however, Formosa is not an El Dorado, it is an island whose future promises to be a bright one. Richly endowed by nature, it is greatly favoured by its insular position. The Chinese, always averse to change, are anxious that Formosa should form an experimental province, and are prepared to remove restrictions there, which for the present they seem inclined to continue on the mainland. Sufficiently supplied as it is with labour at a reasonable rate, and inhabited by a healthy and industrious people, it presents a splendid field for commercial enterprise.

The question has often been asked, Should such an island be left in the hands of such a government as the Chinese? To argue for a moment that the Chinese Goverment is in any respect superior, say to the British, would be ridiculous, but to suggest that

therefore the Chinese should be deprived of part of their possessions is highly improper. Their sovereignty over the island is better established than ours was in New Zealand a very short time ago. They are employing their military in making roads through the savage districts, the most effectual way of eventually subduing them, and they seem anxious to develop the resources of the country. In more than one European newspaper there have been suggestions as to annexation, but any attempt at such a thing would, in my opinion, be nothing short of robbery.

G. JAMES MORRISON.

# Reviews.

## COLONEL MEADOWS TAYLOR.*

In Colonel Meadows Taylor we lost a valued contributor to this magazine, who took a special interest in the various Indian Surveys which it has been one of our principal objects to make more generally known and appreciated. His review of Dr. Hunter's *Orissa* in our number for July 1872, and other contributions, show with what keen interest Colonel Meadows Taylor followed the efforts of those younger men who are working for the good of India by accumulating and classifying accurate information. No one would more thoroughly appreciate any successful attempt to illustrate the life and modes of thought of Eastern people, and he showed this in his warm and cordial welcome of Mr. Palgrave's *Hermann Agha* in our number for September 1872. But his loss is more especially felt by us, because we can never again receive that valuable and weighty advice on matters relating to Indian questions, and especially to revenue surveys and to archæology, from which we have often derived such benefit in recent years.

The story of the life of Meadows Taylor is one which it was right to give to the world, for it teaches many lessons—it will cheer and encourage the young and friendless when the difficulties before them seem too great for them to surmount, it will show them that honours and advancement are within their reach, and it will strengthen them in their resolution to deserve success. Such a story will do more good than any amount of general advice or admonition, for it teaches by example, and proves how certainly a life of steady and resolute perseverance in the path of duty will be rewarded. For Meadows Taylor was not one of those men who have gone to India with many introductions, to be pushed forward by relations and by the influence of patronage. No one could have been more friendless, and indeed unfortunate, at the opening of his career. As a boy of fifteen he was sent out to Bombay in 1823, upon a vague and fallacious promise of employment in a merchant's office. When he arrived, the house of business proved to be no better than a shop, and its chief an embarrassed tradesman. The misfortune-stricken boy found one friend, who obtained for him a commission in the Nizam's contingent. All his future advancement was due entirely to himself. But Meadows Taylor was wholly without self-assertion, and those pushing qualities to which the success of many friendless men is often due. To those who knew him, and to those who read the story of his life, his modesty and single-mindedness are the qualities which are most prominent in his character. He used the talents which God had given him to the utmost of his power for the good of the public service, without one selfish end, and without a thought as to whether he was more or less gifted than his neighbours. Yet he was endowed with rare gifts of imagination and memory, steadied by habits of self-control; and his ability was accompanied with prudence and sound judgment.

No man could have been better suited to fill the extremely difficult post of Resident, and virtually of administrator in a remote and somewhat turbulent native State. In this capacity he worked for many years at Shorapur, in the Deccan; and while maintaining the dignity and superiority which were called for by his position, he ever had a warm sympathy for the people, and treated them with respect as well as kindness, so that he won not only their esteem but their love. No Englishman in India was ever more beloved by the people among whom he worked; and certainly the successful administration of a State like Shorapur, where every branch of government must be undertaken by one unaided man, requires very exceptional qualifications. Colonel Meadows Taylor was an excellent linguist; he acquired a considerable knowledge of law; he studied botany and geology, because he observed the direct bearing of those sciences on the productiveness of the soil; and he taught himself the science of engineering, in order that he might design and execute important works of irrigation. Above all, he was richly endowed with imagination, without which there cannot be a good administrator, especially when dealing with people of a different race. Indeed, it is not too much to say that all his other acquirements would have been little worth, if the gift of imagination had been absent.

Those fascinating pictures of native Indian life with which Meadows Taylor enriched our literature are due to the same faculty which enabled him to advance the moral and material well-being, while he won the love and respect, of the people who were placed under his charge. His earliest work has stood the test of time, and the *Confessions of a Thug*, in which the manners and superstitions of the people of India are portrayed with wonderful fidelity, still retains its reputation. But his later romances, designed to mark the principal epochs of Indian history, have exercised an equally fascinating power over their readers. *Tara* is intended to give a picture of the rise of Mahratta power, and describes, as no history can do, the ways of life and modes of thought of the different races of the Deccan two centuries ago, and vividly brings home to the reader not only great historical events, but the scenery amidst which they were enacted. *Ralph Darnell* portrays the events in the days of Holwell and Clive in Bengal; *Tippoo Sultaun* introduces us to the great events in Southern India in the time of Cornwallis; while *Seeta* treats of the terrible period of the mutinies in 1857. They form a complete work in themselves, and Mr. Reeve

* *The Story of My Life.* By the late Colonel Meadows Taylor, Author of "Confessions of a Thug," "Tara, a Mahratta Tale," &c. Edited by his Daughter, with a Preface by Henry Reeve. Two vols. (Blackwood, 1877.)